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22.1 Basics

Taping the hand and fingers for crack climbing has long been a commonplace practice to prevent skin injuries. Professional boxers, football players, and basketball players often use tape to prevent injuries and to allow an early return to competition after an injury. Taping can help an athlete to work at his sport with some protection while not completely immobilizing the injured area as with a cast. Using tape correctly requires a proper diagnosis of the injury and the correct application of the tape. Several procedures used as treatment may also be used to prevent injury or to support recovery [1]. In the following chapter, we will focus on therapeutic, static taping. Dynamic Kinesio® Taping (k-taping), which has a completely different ratio and approach, is used less frequently in rock climbing and will therefore not be included. Protective taping of the skin for crack climbing or for some finger pockets will help to reduce pain and skin damage; nevertheless, general prophylactic taping is not recommended. Surprisingly, a study by Woollings et al. [1] even highlighted it to be a risk factor for finger injuries! Nevertheless, in the rehabilitation after finger injuries, it is an essential tool when returning to climbing. So far, little scientific work has been performed concerning taping and climbing [2–9]; thus, some of the information in this chapter may have only level 4 evidence. Nevertheless, we try to present this topic with as much scientific evidence as possible.

Taping is used in climbing for various reasons:

1. A preventative (protective) taping against skin abrasions and injuries is performed in crack climbing as well as for certain finger pockets. In certain, repetitive work on one hard move in a climb, a preventive joint capsule tape may be used [10]. A general prophylactic finger taping is not recom-

mended and was even found to increase injury incidence [1].

2. A therapeutic taping represents one part of therapy for (finger) overstrain injuries or during return to (climbing) sports after an acute (finger) injury. Also, some acute finger injuries (e.g., collateral ligament injuries) can receive therapeutic tapings for partial immobilization.

Applying static tapings in climbing should follow certain guidelines as insufficient taping can be ineffective or even harmful.

General Taping Guidelines [10]:

- Tape should never be applied to the skin directly off the roll. To tape correctly, a strip of tape should be torn off at the desired length and then carefully applied to the injured area. Taping an injury straight from the roll usually makes the application too tight and causes skin breaks.
- The width of the tape must relate to the size of the finger. It is crucial to apply the correct tension while applying the tape. This being said, it is also necessary to create soft tissue “windows.” Thus, the subcutaneous tissue can release in these areas without being excessively compressed underneath the tape. A finger that is fully covered underneath one or several layers of tape is not advisable and harmful.
- As a general principle, taping should not be circular around a limb, as to not compromise blood flow. The fingers are an exception because, otherwise, the tape would not be stable enough and would loosen while climbing. Nevertheless, the tape should not restrict the blood flow or damage the finger nerves.
- During the application of the tape, the finger must be in a functional position, depending on what effect the tape should accomplish. For example, if a complete extension of the PIP joint should be prohibited, as in the case of a joint capsule injury, the tape must be applied in a 30° flexion of the PIP joint.

- The respective body region should not be shaved directly before taping, as this may cause micro skin abrasions which can cause infections.
- The skin should be dry and clean before taping.

22.2 Pulley Taping (Pulley Injury, Tenosynovitis) (H-taping)

Indication: Pulley injury, Tenosynovitis

After closed pulley injuries in climbers, a specific technique of finger taping is recommended to decrease the increased tendon bone distance (bowstring). Originally, circular taping around the base of the phalanx was recommended [7, 11]. Schweizer et al. [8] determined a more distal and oblique tape position to be more effective. Further biomechanical studies by Roloff et al. [12] lead to the invention of an even more effective technique, the H-tape [6]. This effectiveness of this technique was confirmed by a cadaver study in which human fingers were loaded with various tapings in various grip positions [6]. The new taping method decreased the tendon bone distance in the injured finger significantly by 16% and was superior to other taping methods. The strength development was significantly better with the new tape for the crimp grip position (+13%), while there was no significant improvement for the hanging position [6]. The study could not observe any increase in strength in the healthy finger after having applied the tape, which undermines the observations of Schweizer et al. [8] and Warme and Brooks [7] that taping can never approximate or improve upon the effectiveness of an intact pulley system [6].

This taping technique is also effective in the therapy of tenosynovitis of the finger flexor tendons as it decreases the deflection of the tendons at the distal rim of the A2 and the proximal rim of the A4 pulley, thus decreasing friction [13, 14].

In a pulley injury, the flexor tendons experience an increased distance to the bone while flexing the finger. This is called the “bowstring phenomenon,” and it increases with the grade of finger flexion. The forces acting on the pulley are strongest in a

flexed position. The force being applied on the finger flexor tendon pulley system is a function of the tension being developed in the flexor tendons and the angle between the tendon and the pulley [12]. In order to decrease the force on the pulley (potentially causing the rupture), the goal must be to diminish this angle since the tendon tension cannot be influenced [12]. While the A2 and A4 pulleys are rather stiff, the A3 pulley is weak, thin, and very flexible as it is attached on the palmar plate, which increases its flexibility during motion, as shown by an MRI study by Bayer et al. [15]. In an A2 or A4 injury, the flexor tendons separate from the bone (bowstring) as the A3 stretches and do not influence the path taken by the tendon.

A biomechanically effective tape thus needs to counteract the bowstring of the tendon, which is most effective at the point of the most pronounced distance between the tendon and bone. The maximum distance is situated at the point of maximum joint flexion of the PIP which coincides with the A3 pulley and on the outside, the crease in the skin over the PIP joint on the palmar aspect of the finger. Additionally, at the level of the pulleys, there is a substantial amount of subcutaneous tissue, whereas at the level of the flexor crease at the PIP joint, this is minimal. In consequence, tape applied over the crease of the skin at the PIP joint can deliver maximum support to the flexor tendons in comparison to other anatomical landmarks. After the initial study and invention of the H-tape, multiple variations have been discussed online. However, only the original technique has been investigated scientifically so far [10].

For the application of a correct H-tape, a 1.5 cm-wide and 8–10 cm-long strip of tape is cut (size varies according to finger size). The strip is then split longitudinally from both ends in such a way that a central “bridge” of 1 cm width remains (branches ~0.75 cm wide, Fig. 22.1) [13]. The central “bridge” of the tape is pressed onto the flexor fold of the PIP joint. The two proximal branches are then applied rather tightly onto the base phalanx. With the PIP joint flexed at about 45–60°, the two distal branches are then tightened over the middle phalanx (Fig. 22.2). The tape should be rather tight; thus, it should only be applied directly before climbing to minimize



Fig. 22.1 The tape is split from both ends, and the branches should not be too wide. Normally, we use a 1.5 cm-wide and 8–10 cm-long strip of tape; thus, the branches are about 0.75 cm wide each (Photo: Michael Simon)



Fig. 22.2 Applying the tape: the PIP joint is flexed, and first the distal branches and then the proximal branches are fixed (Photo: Michael Simon)



Fig. 22.3 The bridge of the tape is directly upon the flexor fold of the PIP joint (Photo: Michael Simon)

potential blood flow restriction (Fig. 22.3). It is crucial to know that the tape will loosen during climbing. To enhance the tape, a second layer (a

thin figure-eight-shaped tape) can be applied on top of the H-tape to protect the base-layer [10].

1. As the tape is exactly positioned over the finger middle joints folding crease, it accomplishes various effects. The finger's subcutaneous fat and tissue is minimal at this point; the tape's pressure can directly work on the force transmission of the flexor tendons.
2. The tape supports the A3 pulley. The A3 pulley is thinner and weaker, thus more flexible than the A2 and A4 pulleys. Additionally, it is not attached directly to the bone but to the palmar plate, which increases its flexibility during motion.
3. The tape indirectly decreases the forces acting on the injured A2 or A4 pulley [13] (Chap. 6.3.1. Figs. 11 and 12).

The effect of the H-tape also works similarly for tenosynovitis. During the crimping grip position, the flexor tendons are bent at the distal end of the A2 pulley and at the proximal end of the A4 pulley [16, 17]. This leads to an increased flexion of the tendons and thus increased friction on the rim of the pulley [13, 17]. Repetition of this position can lead to a chronic inflammatory reaction of the tendon sheath [18]. As the H-tape decreases the angle of flexor tendon deflection, friction will be reduced, and the tenosynovitis can recover [14, 18]. In an ultrasound study, Schöffl et al. [6] showed that only the H-tape can significantly decrease the tendon bone distance and increase finger strength taping after finger pulley rupture. No benefit was found for any tape in an intact pulley system. After a pulley rupture, taping during climbing should be performed for a few months (see Chap. 6 "Finger Injuries"). This helps to reduce the risk that symptoms will become chronic or an onset of tenosynovitis based on finger pulley injury. While the H-tape is found to work effectively in tenosynovitis [13, 14], circular tape at the base of the proximal phalanx can also be effective in cases of minor tenosynovitis [10]. Alternatively, a more distal taping method as shown by Schweizer et al. [8] can be used for both pathologies.

22.3 Finger Middle Joint (PIP Joint)

Indication: Middle joint capsule injury, capsulitis, strain of the flexor tendons, and lumbricals

The proximal interphalangeal joint can be best supported by a “figure eight” (or 8-shaped) tape. It is important to apply it in a functional finger position, meaning a 45° flexion of the PIP joint, so the tape reduces the overall extension, while the full finger flexion is still attainable. Fist closure, rope and belay device management, and other hand functionalities are guaranteed [10]. The finger middle joint’s capsule undergoes the most stress in passive overextension. The same applies to the flexor tendons, which are sensitive to overextension. Thus, the aim of the tape is to avoid finger hyperextension, which is why it needs to be applied in a flexed finger position. If the tape were to be applied in an extended finger position, it would be ineffective.

Starting at the base of the finger, the first wrap of tape should go around the base phalanx, as close to the palm as possible. While crossing the middle joint, it is important to keep 30–45° of flexion in this joint. The tape then encircles the middle phalanx and comes back to the base phalanx, crossing palmar sided at the PIP joint. At this point, the 30–45° flexion is highly important. The tape is finished by encircling the base phalanx. Similar to the H-tape, it is important that the tape branches are not too wide, enabling stress to be applied to certain areas, while other areas work as stress release for (sub-)cutaneous tissue. Therefore, a soft tissue window should be kept free of tape on the palmar side of the proximal joint. As previously stated, full coverage of the finger with tape is insufficient and harmful [10] (Figs. 22.4, 22.5, 22.6, 22.7, and 22.8).

22.4 Finger Middle Joint Collateral Ligament Injury

Indication: Collateral Ligament Injury PIP

A collateral ligament injury to the PIP joint can also be effectively supported by a modified figure eight tape. In the initial phase of the injury,



Fig. 22.4 Taping of the finger middle joint: starting at the base of the finger, the first wrap of tape should go around the phalanx as close to the palm as possible (Photo: Michael Simon)



Fig. 22.5 While crossing the middle joint, it is important to keep 30–45° of flexion in this joint (Photo: Michael Simon)



Fig. 22.6 The tape circles around the middle phalanx (Photo: Michael Simon)

a splint immobilization is advised followed by a buddy-taping [18]. In the next phase of recovery, a modified figure eight tape is used. Therefore, the crossing of the eight is not applied to the palmar side at the PIP joint but onto the injured lateral side. It is highly important that the finger is



Fig. 22.7 It then crosses again palmar sided the PIP joint (Photo: Michael Simon)



Fig. 22.8 A soft tissue release window is important (Photo: Michael Simon)

flexed at 30–45°, while the lateral crossings are performed. The collateral ligaments of the fingers are mostly tensioned in a slight flexion of about 30°; while in a complete extension, the PIP joint has some bony lockage. Alternatively, various single straps can be taped out by doing a base circulation at the finger's middle and base phalanx combined with longitudinal straps based on top of both sides which are finally fixed by another layer of circulations at the base and middle phalanx. As this tape consists of multiple layers, it tends to protrude more and also tears off more easily. We therefore recommend the lateral-sided figure eight [10] (Figs. 22.9, 22.10, 22.11, and 22.12).

22.5 Finger End Joint (DIP Joint)

Indication: Joint capsular injury, capsulitis, ganglion cyst



Fig. 22.9 In a collateral ligament injury, the tape starts as with a classic figure of eight tape close to the fingers base but crosses the PIP joint in 30–45° flexion at the respective side of the injured ligament (Photo: Michael Simon)



Fig. 22.10 After circling around the middle phalanx, the tape is lead back to the base phalanx, crossing the PIP joint sideways, which is in a 30–45° flexion (Photo: Michael Simon)



Fig. 22.11 The tapes' crossing is sideways at the PIP joint (Photo: Michael Simon)

As the distal interphalangeal joint (DIP) is tiny, a figure eight taping does not work there. Additionally, it would cover too much of the fin-

gertip and decrease friction while climbing. Thus, for the finger end joint, we recommend plain circular tape directly around the joint [10] (Figs. 22.13 and 22.14).



Fig. 22.12 The tapes' crossing is sideways at the PIP joint (Photo: Michael Simon)



Fig. 22.13 To stabilize the DIP joint, it is directly encircled by a thin tape layer (Photo: Michael Simon)



Fig. 22.14 To stabilize the DIP joint, it is directly encircled by a thin tape layer (Photo: Michael Simon)

22.6 Finger Base Joint (MCP Joint)

Indication: Joint capsular injury, capsulitis, ligament injuries, injuries to the connexus intertendineus of the extensor tendons

Taping of the finger base joint (MCP joint) is a bit trickier but can be effective. For this taping technique, the skin must be dry and clean, as the tape has free ends. During its application, the MCP joint is flexed at about 20°. The branches of the tape should cross each other, one distal and the other proximal to the knuckle (head of the metacarpal). The ends of the tape are only pressed onto the dorsal aspect of the hand, without any additional reinforcement (tape sticks better to tape than to skin). We found that these free ends still hold onto the skin well and do not come off too easily [10] (Figs. 22.15 and 22.16).



Fig. 22.15 Tape stabilization of the MCP joint needs to be performed in 30° flexion with the tape branches crossing each another. These crossings are distal and proximal of the “knuckle” (Photo: Michael Simon)



Fig. 22.16 Tape stabilization of the MCP joint needs to be performed in 30° flexion with the tape branches crossing each another. These crossings are distal and proximal of the “knuckle” (Photo: Michael Simon)



Fig. 22.17 One tape fixes the base phalanx, and the other fixes the middle phalanx to the adjunct finger. If the lengths of the phalanges are very different, these strips must be attached obliquely (Photo: Michael Simon)



Fig. 22.18 The tape starts at the index finger and runs dorsally behind the (injured) middle finger to the ring finger (Photo: Michael Simon)

22.7 Buddy-Taping

Indication: Collateral ligament injury PIP, joint capsular injury, lumbrical injury

In severe PIP joint capsular injuries, collateral ligament injuries, and lumbrical muscle tears, buddy-taping is very effective for both therapy and during the process of returning to sport. The healthy finger thereby works as the “buddy,” stabilizing the injured finger [10] (Fig. 22.17).



Fig. 22.19 After encircling the ring finger, the tape comes back to the extensor side of the ring finger and is attached to itself (Photo: Michael Simon)

22.8 Lumbrical Tape

Indication: Lumbrical muscle strain or tear

Lumbrical muscle injuries can be very effectively treated with taping [19]. In the acute phase, a “buddy”-tape is used; later, we recommend the lumbrical tape, as shown here. The tape is applied dorsally to the injured finger. This allows the finger to freely bend but prevents the adjoint fingers from flexing far if the injured finger is pulling in an extended position. Thus, the lumbrical muscle can be relieved from stress (Figs. 22.18, 22.19, and 22.20). This taping technique is also very helpful in restarting climbing after a lumbrical strain.



Fig. 22.20 The tape then ends at the index finger and prevents the adjunct fingers from being bent far in a mono-pull of the middle finger. Thus, the lumbrical muscles are relieved from stress (Photo: Michael Simon)

22.9 Wrist

Indication: Tenosynovitis, discus lesion, synovitis in the ulnocarpal recesses

With the increasing number of people who have become interested in climbing, climbing-

related injuries of the wrist and the forearm are currently seen more frequently than they have been in the past. As more and more athletes perform the sport on a level which had previously only been reached by a few top athletes, injuries such as hamate fractures or other pathologies of the wrist have become more common [20–22]. Circular wrist taping is a technique commonly used by athletes in different sports to both strengthen and stabilize the wrist for a high-compression load (e.g., weightlifting [23, 24], gymnastics [25, 26]) or to stabilize the wrist for tensile stress as present in climbing when the athlete is pulling on his/her hand [10, 20]. However, in their 1997 study, Rettig et al. proved that circular wrist tape, with or without additional taping of the fingers, does not increase wrist strength [9]. Contrary to the perceptions of the athletes, Takahashi et al. even showed that tight circular wrist tape can even slightly decrease grip strength when applied too tightly. However, no climbing-specific data on grip strength variations under wrist tape has been published so far.

Climbers suffering from a feeling of instability in the wrist or nonspecific wrist pain are often diagnosed with injuries of the ligaments and capsules or even with bone marrow edema of the carpal bones or carpal fractures [10, 20]. The pain and discomfort mainly arise in radial/ulnar abducted positions of the hand or while grasping an undercling (position of the hand in maximum supination) [10, 20].

During climbing or bouldering, the majority of thenar and hypothenar muscular strength is transferred to the carpal bones and the distal part of the radius by the transverse carpal ligament, causing frequent reactions of the lunate and the distal radius [22, 28]. If maximum strength is applied in a slightly ulnar abducted and dorsally flexed position of the hand, the carpal bones (especially the lunate) are in a relatively unstable position, as biomechanical analyses have shown [4, 28]. Lutter et al. [20] published a study which showed that high stress to the flexor tendons can even cause fractures of the carpal bones. It is comprehensible that radioulnar joint instability or anatomic predispositions such as incongruent wrists

(pos./neg. ulnar variance) can exacerbate the development of problems in this region [4, 22].

22.9.1 Taping Technique

Unlike finger tapes, for which thin strips of tapes are used [6], wider tapes are recommended for circular wrist tapes [4]. As climbers need both stability within the wrist and a high level of flexibility, the tape strips are recommended to be approximately 2.5 cm wide. This width allows for an increase of pressure within the carpal while maintaining the flexibility of the wrist. Due to the fact that this taping technique increases the carpal and thus adds stability within the wrist and not the distal part of the forearm, it is recommended to place the tape distally to the styloid process of the ulna (skin fold of the wrist) [4] (Fig. 22.21). This supports the carpal architecture, including the transverse carpal ligament. In contrast to Takahashi et al. and Rettig et al., who investigated athletes directly after taping the wrist, we recommend applying the circular wrist tape rather firmly (two to three layers) due to the fact that even non-elastic tape loosens a little bit within the first minutes of climbing [4, 9, 10]. Additionally, in mantle and stemming moves, the climber applies high pressure to the distal radio-ulnar joint, which allows some grade of divergent movement. If the tape is too tight, this would be painful. It must be considered that not all wrist conditions benefit from circular taping, which



Fig. 22.21 The wrist is encircled one and a half to two times with a 2 cm-wide tape, as far distally as possible (Photo: Michael Simon)

may be explained by the intra-articular increase of pressure due to the tape [4]. If the climber's pain worsens with the tape, it should not be continued [4].

22.10 Base Joint of the Thumb

Indication: Joint capsule and collateral ligament injuries, hypermobility

Strains or tears of the ulnar collateral ligament of the thumb, commonly suffered in falls while skiing, can cause a severe pain in the respective area. If a serious injury or complications (e.g., Stener lesion) are excluded, taping the thumb can help reduce pain and offer stabilization and protection. The tape starts with two base layers around the wrist and the proximal phalanx of the thumb. The following layers are then applied like roof tile, with one layer stacking over the previous one. The crossing point of the tapes should be at the point of the injury; thus, the tape can be varied accordingly. This technique creates support for the thumb similar to a soft splint (Figs. 22.22, 22.23, and 22.24) [10].



Fig. 22.22 The tape branches will be set so they cross each other at the dorsum, starting distally and continuing stepwise proximally, partially overlapping (Photo: Michael Simon)



Fig. 22.23 This is continued proximally (Photo: Michael Simon)

22.11 Hand Taping in Crack Climbing

Indication: Prophylactic in crack climbing

There are a number of products that help protect the hands when crack climbing. The traditional form of protection is athletic tape; we demonstrate a simple “one-strap” method to produce a “crack glove” [10]. There are more complicated methods of taping around the fingers; however, this method is effective and simple. This is one of the few methods in which the tape is attached to the hand directly off from the roll. The tape is pulled out from the roll in 20–30 cm increments before being wrapped around the hand. The tape should not be pulled while wrapping, as this will inadvertently make the glove too tight.

To start, the end of the tape is placed on the back of the wrist with the fingers spread as wide as possible. The tape glove is started with one to



Fig. 22.24 Finally, an anchoring strip is placed on top horizontally (Photo: Michael Simon)

two circular wraps around the wrist and then continued diagonally across the back of the hand. At the base of the finger, the tape makes a slight turn, the hand is turned over, and the tape is placed at the very end of the palm. The tape is then brought

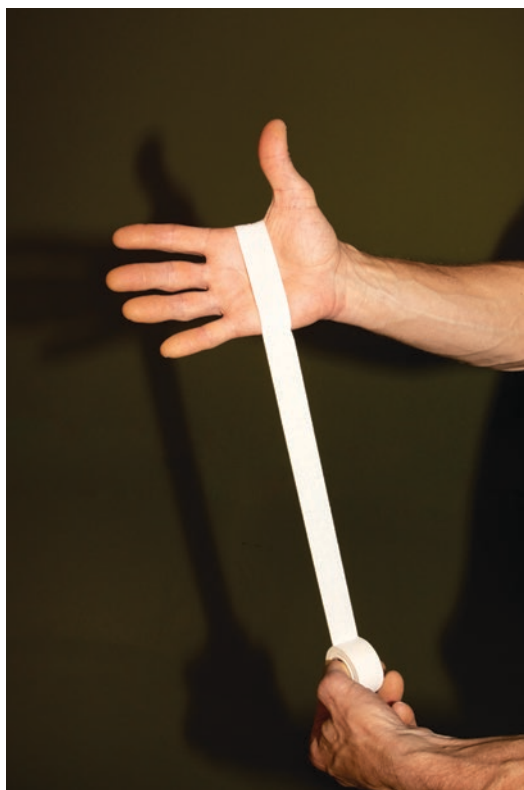


Fig. 22.25 The tape starts with one to two circulations around the wrist and then runs dorsally diagonally over the back of the hand. Then, it wraps around the palm twice and comes back diagonally over the back of the hand to the wrist, where it is wrapped around again (Photo: Michael Simon)



Fig. 22.26 The tape starts with one to two circulations around the wrist and then runs dorsally diagonally over the back of the hand. Then, it wraps around the palm twice and comes back diagonally over the back of the hand to the wrist, where it is wrapped around again (Photo: Michael Simon)

back to the dorsal aspect of the hand, placed just over the MCP joints and again across the same area of the palm. Rounding the blade of the hand, the tape is pulled toward the wrist. Another wrap is performed around the wrist, finishing the glove. This method covers part of the palm, but that area usually separates from the tape while climbing, allowing the tape to roll into a wide strip that is independent of the palm (Figs. 22.25, 22.26, 22.27, 22.28, 22.29, and 22.30) [10].

22.12 Elbow Taping

Indication: Epicondylitis, hypermobility of the ulnar nerve, brachialis tendinosis

Taping the elbow is used as therapy and prophylaxis for epicondylitis (medial or lateral), bra-

chialis tendinosis, and an irritated, hypermobile ulnar nerve. As many larger taping techniques and external braces reduce venous blood back-flow and increase the feeling of “pumped” forearms, we use a singular tape directly over the epicondyles at the elbow’s flexor fold [10]. This technique is applied by wrapping a 2–2.5 cm-wide strip of tape around the elbow. It is important to flex the elbow at 90° and tense the biceps when closing the tape. With elbow flexion and biceps tension, the diameter of the elbow increases; tape applied onto a relaxed muscle in extension would cause an excessively tight tape that is prone to rupture. Correctly applied tape should be rather loose and lay in folds on the skin. When the elbow is flexed and the bicep muscle is tensed, pressure is applied to the epicondyles. For this taping technique, the effective



Fig. 22.27 The tape starts with one to two circulations around the wrist and then runs dorsally diagonally over the back of the hand. Then, it wraps around the palm twice and comes back diagonally over the back of the hand to the wrist, where it is wrapped around again (Photo: Michael Simon)



Fig. 22.28 The tape starts with one to two circulations around the wrist and then runs dorsally diagonally over the back of the hand. Then, it wraps around the palm twice and comes back diagonally over the back of the hand to the wrist, where it is wrapped around again (Photo: Michael Simon)

mechanism is more through stimulation of the proprioceptors than a pure mechanical effect. Overall, this taping technique is more accommodating during climbing than external braces, as these decrease blood flow and thus increase “pumped” forearms (Figs. 22.31, 22.32, 22.33, and 22.34) [10].

22.13 Taping of the Ankle

Indication: Sprains, chronic instability

Ankle injuries (strains and sprains) are frequent, especially in bouldering, and supportive taping can be used as therapy, depending on the extent of the injury, as well as during the process of returning to sport. There are many varia-

tions of ankle-joint tapings presented in literature. We demonstrate an easy and reproducible technique.

First, a base layer (anchor strips) of two semicircular straps is applied at the forefoot and the lower leg. Then, the actual holding strips, which are “U”-shaped (stirrup), are applied to the front and to the lower leg. There should be three strips of each, applied alternately. The most important strip is then applied, fixing the foot in a pronation through elevation of the lateral foot edge. While this being applied, it is crucial that the foot is kept in a 90° flexion and pronation. Finally, the tape can be covered with optional semicircular sheeting. We would definitely apply the sheeting to severe injuries (following the acute phase), but



Fig. 22.29 The tape starts with one to two circulations around the wrist and then runs dorsally diagonally over the back of the hand. Then, it wraps around the palm twice and comes back diagonally over the back of the hand to the wrist, where it is wrapped around again (Photo: Michael Simon)



Fig. 22.30 The tape starts with one to two circulations around the wrist and then runs dorsally diagonally over the back of the hand. Then, it wraps around the palm twice and comes back diagonally over the back of the hand to the wrist, where it is wrapped around again (Photo: Michael Simon)

if the tape is supposed to support the foot in a climbing shoe, it may be too thick. It must be mentioned that this taping technique is circular and cannot be used directly after an acute trauma, as this may cause compartment syndrome. After completion of this taping technique, the distal tape is cut from distal to proximal for about 1–1.5 cm in the first interdigital space. In a normal standing position, the metatarsals yield transversally to the weight, and the actual diameter of the forefoot increases; thus, the tape would be too tight (Figs. 22.35, 22.36, 22.37, 22.38, 22.39, 22.40, 22.41, 22.42, 22.43, 22.44, 22.45, 22.46, and 22.47).

22.14 Conclusion

Taping techniques, especially for the hand and fingers, are important therapeutic tools for climbers, especially during rehabilitation and the process of returning to. Specific biomechanical tapes can relieve stress on the finger flexor tendon pulleys, the tendons, and the joint capsule. Correct application of these tapings is essential, and soft tissue “windows” are important in order to ensure that a certain pressure can be applied to various areas, while other areas release the (sub-)cutaneous tissue. Fully covering a finger with tape is ineffective and harmful.



Fig. 22.31 The tape runs directly over the ulnar epicondyle, goes around the back of the arm, and comes to the lateral epicondyle. It is then closed at the front. For closure, the biceps must be tensed and the tape must be applied very firmly. In a relaxed arm, the tape should have folds which these unfold as the tape is under tension during muscle activation and flexion. Thus, it works with direct pressure onto the epicondyles and muscle insertions while guaranteeing free blood flow when the arm is relaxed (Photo: Michael Simon)



Fig. 22.32 The tape runs directly over the ulnar epicondyle, goes around the back of the arm, and comes to the lateral epicondyle. It is then closed at the front. For closure, the biceps must be tensed and the tape must be applied very firmly. In a relaxed arm, the tape should have folds which these unfold as the tape is under tension during muscle activation and flexion. Thus, it works with direct pressure onto the epicondyles and muscle insertions while guaranteeing free blood flow when the arm is relaxed (Photo: Michael Simon)



Fig. 22.33 The tape runs directly over the ulnar epicondyle, goes around the back of the arm, and comes to the lateral epicondyle. It is then closed at the front. For closure, the biceps must be tensed and the tape must be applied very firmly. In a relaxed arm, the tape should have folds which these unfold as the tape is under tension during muscle activation and flexion. Thus, it works with direct pressure onto the epicondyles and muscle insertions while guaranteeing free blood flow when the arm is relaxed (Photo: Michael Simon)



Fig. 22.34 The tape runs directly over the ulnar epicondyle, goes around the back of the arm, and comes to the lateral epicondyle. It is then closed at the front. For closure, the biceps must be tensed and the tape must be applied very firmly. In a relaxed arm, the tape should have folds which these unfold as the tape is under tension during muscle activation and flexion. Thus, it works with direct pressure onto the epicondyles and muscle insertions while guaranteeing free blood flow when the arm is relaxed (Photo: Michael Simon)



Fig. 22.35 A base layer with an adhesive bandage is applied to avoid skin alterations (Photo: Michael Simon)



Fig. 22.37 Proximal and distal tape anchors are fixed in two half strips (Photo: Michael Simon)



Fig. 22.38 Proximal and distal tape anchors are fixed in two half strips (Photo: Michael Simon)



Fig. 22.36 A base layer with an adhesive bandage is applied to avoid skin alterations (Photo: Michael Simon)



Fig. 22.39 The first retaining strap is applied in a “U”-shape (stirrup). It starts at the plantar aspect and is then attached at the lower leg. In the next step, the rest of the tape is molded onto the base layer with the palms (Photo: Michael Simon)



Fig. 22.40 The second retaining strap comes dorsally to secure the talus. Three of these two respective strips are applied, alternating and partially overlapping each another (Photo: Michael Simon)



Fig. 22.43 The tape starts medially proximally (Photo: Michael Simon)



Fig. 22.41 It is crucial that the foot is bent at 90° and is in pronation during the taping (Photo: Michael Simon)



Fig. 22.44 The tape runs around the lateral side of the foot to work as a pronation strap and is continued to the cranial aspect (Photo: Michael Simon)



Fig. 22.42 The next strap is the most important for stability and runs obliquely underneath the medial malleolus (Photo: Michael Simon)



Fig. 22.45 Optional sheeting is applied with half wraps (Photo: Michael Simon)



Fig. 22.46 Optional sheeting is applied with half wraps (Photo: Michael Simon)



Fig. 22.47 Finally, the tape is cut between the first and the second phalanges to ensure a spread within the forefoot (Photo: Michael Simon)

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